

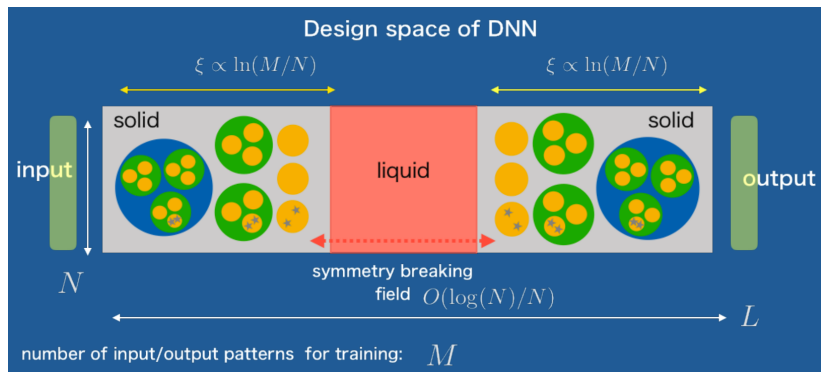
Depth-controlled glass transition in deep neural networks

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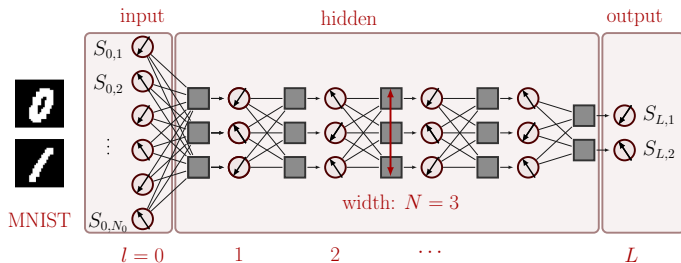
The behavior of deep neural networks (DNNs) is puzzling!



The design space of DNN based on mean field replica theory (Hajime Yoshino, 2020).

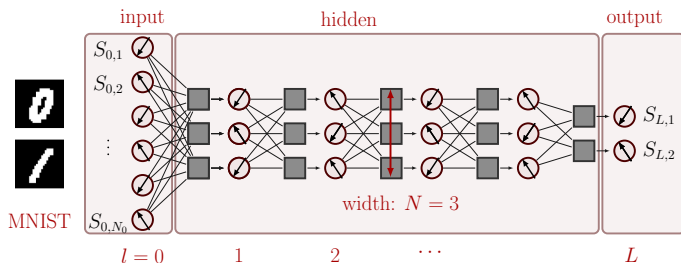
- **DNNs**: fast; good generalization.
- **Mean field replica theory** predicts: **liquid-to-glass transition** exists in DNNs.

The DNN model (1): Monte Carlo Simulations



- M samples; **Constraint density**: $\alpha = M/N$. (E. Gardner, 1987)
- Activation function $S_{\square}^{\mu} = \text{sgn} \left(\sum_{i=1}^N J_{\square}^i S_{\square(i)}^{\mu} / \sqrt{N} \right)$.
- Hamiltonian $H = \sum_{\mu=1}^M \sum_{\square} V(r_{\square}^{\mu}) = r_{\square}^{\mu 2} \Theta(-r_{\square}^{\mu})$,
 $r_{\square}^{\mu} = S_{\square}^{\mu} \sum_{i=1}^N J_{\square}^i S_{\square(i)}^{\mu} / \sqrt{N}$. $\Theta(x)$: Heaviside step function.
- Multiple replicas, Metropolis Monte Carlo (MC) simulations: update $S_{\square}^{\mu}$ and $J_{\square}^i$ ($i = 1, \dots, N$; $\mu = 1, \dots, M$).

The DNN model (2): Molecular Dynamics Simulations

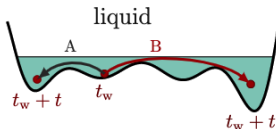


- $v_i(t + \Delta t/2) = v_i(t - \Delta t/2) + F_i(t) \frac{\Delta t}{m_i};$
 $r_i(t + \Delta t) = r_i(t) + v_i(t + \Delta t/2) \Delta t.$
- Force $F_i(t) = -\frac{dE(r_1, r_2, \dots; t)}{dr_i} - \eta v_i(t - \Delta t/2) + \sqrt{\eta(2 - \eta)} k_B T \delta_i.$
- The spins $S_{l,i}^\mu$ are real numbers within the range $-1 < S_{l,i}^\mu < 1.$ Similarly, $-1 < J_{l,i,j} < 1.$ Reflective boundary conditions.
- Define potential of layer l : $V_l = \sum_i \sum_\mu [S_{l,i}^\mu - \tanh(\sum_j J_{l,i,j} S_{l-1,j}^\mu)]^2.$ The total potential $E = \sum_l V_l.$

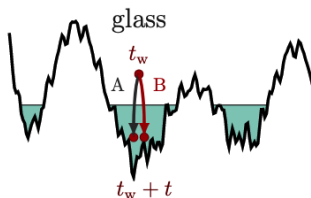
Time correlation functions

Replica-correlation function (for S in layer l)

$$q_l(t, t_w) \equiv \frac{1}{N_l M} \sum_{i=1}^{N_l} \sum_{\mu=1}^M \left\langle S_{l,i}^{\mu,A}(t + t_w) S_{l,i}^{\mu,B}(t + t_w) \right\rangle_{b,r}$$



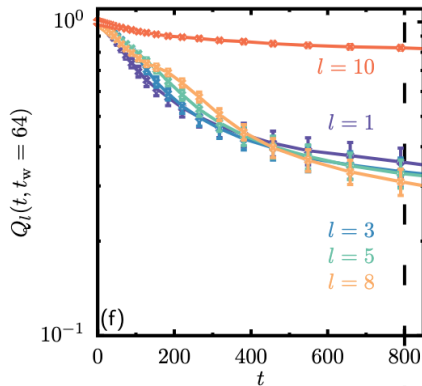
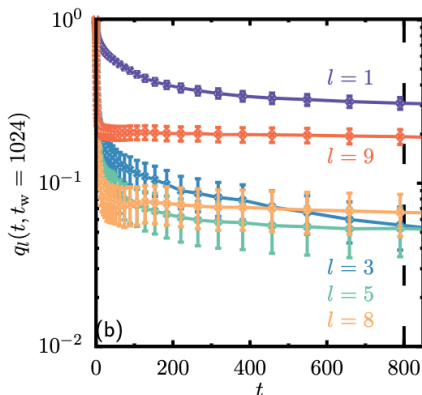
Liquid: relax faster.



Glass: relax slower.

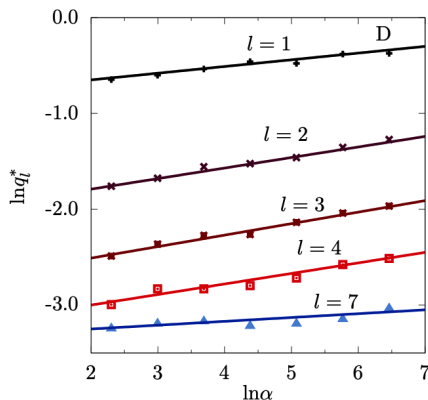
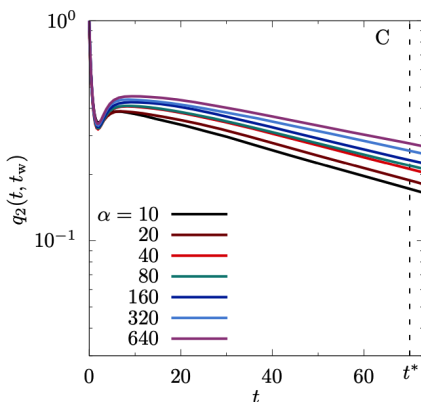
Layer-dependence of the correlation function (from MC)

- In both $q_l(t)$ and $Q_l(t)$, layers near the **boundary** relax **slower**.



Constraint-dependence of the correlation function (from MC)

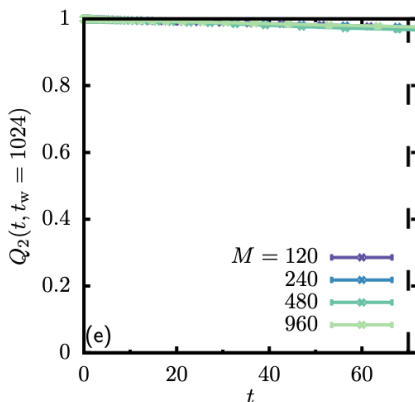
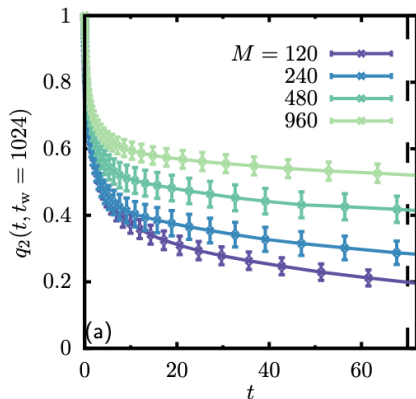
- **Larger** constraint density ($\alpha = M/N$): relax **slower**.
 $(t_w = 1024; q_l^* \equiv q_l(t = t^*, t_w))$



Constraint-dependence of overlaps (2) from MC

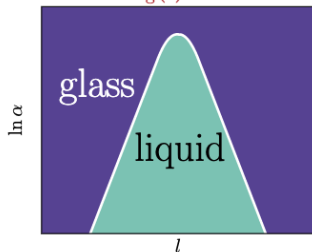
- **Larger constraint density ($\alpha = M/N$): relax **slower**.**

($t_w = 1024$; $q_l^* \equiv q_l(t = t^*, t_w)$)



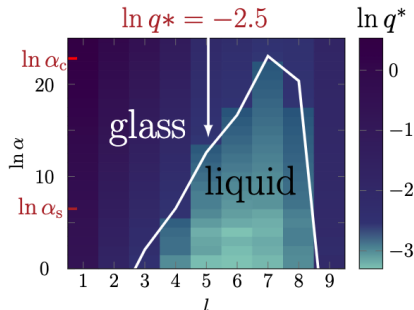
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857



- **Mean field approximation: no loops.**
- **Random training data.**

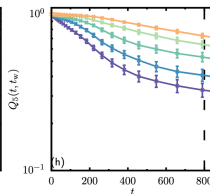
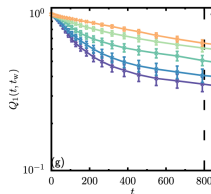
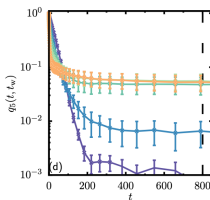
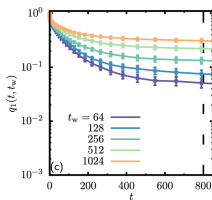
[illegible]



- Real dataset: MNIST.
- Critical α_c exists: $\alpha > \alpha_c$, all layers are in glass phase.

Aging effect in glass phase (2)

- **Glass-phase Layers:** aging effect, **non-equilibrium**. ($l = 1$)
- **Liquid-phase layers:** relax to **equilibrium** quickly. ($l = 5$)
($q'_l \equiv q_l(t = 800, t_w)$)



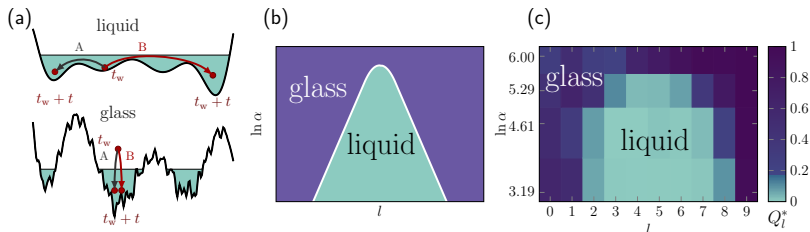


Figure: (a) Loss landscape of liquid & glass phases Two replicated configs A and B decorrelate rapidly in liquids, and are confined in a meta-stable basin in glasses. (b) Phase diagram for the state of network parameters in the l -th layer with a given capacity ratio α , constructed according to the theory $\ln \alpha_g(l) \sim l$. (c) Numerical phase diagram obtained by MD simulations. The liquid-glass boundary is defined at $Q_l^* \approx 0.2$.

Testset accuracy $A(t)$ (MD)

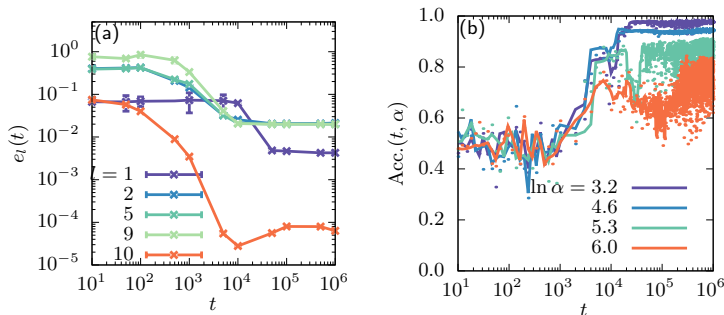
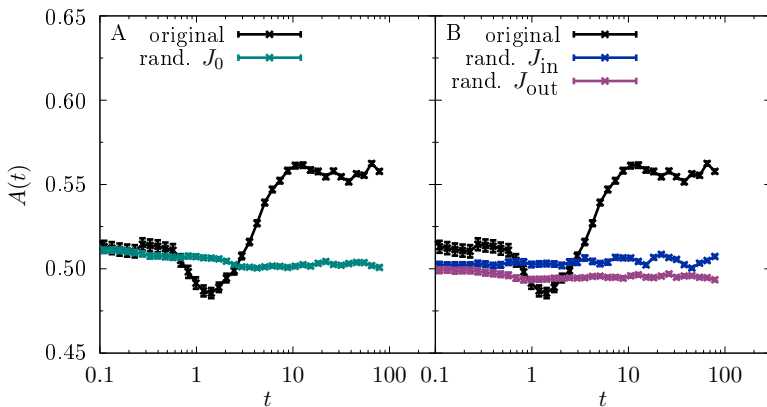


Figure: **(A)** Potential energy density $e_l(t) = V_l/N_l$ during the MD simulation ($\ln \alpha = 5.29$). **(B)** Accuracy on testset decreases as α increases.

Testset accuracy $A(t)$ for DNN (MC)

- After **randomizing** (A) hidden bonds J_0 , and (B) input/output bonds J_{in} or J_{out} between replicas, the accuracy $A(t' = 100)$ **decreases**.



$100 \times M$ boundary conditions; $L = 3$, $\alpha = 40$, $N_{\text{out}} = 1$, $\beta' = 0$.

Testing: the distances of DNN's local minimum (MD)

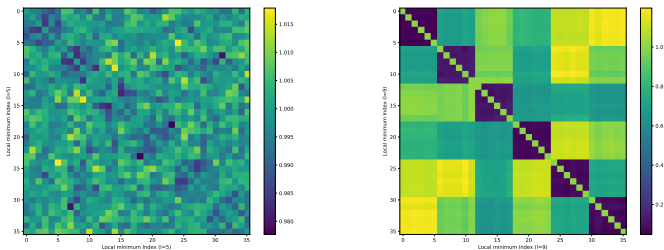
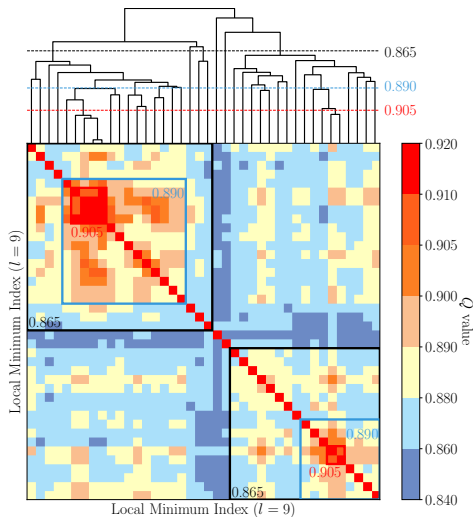


Figure: **(A)** The distances for hidden layer **(B)** The distances from the output layer.

Testing with more initial configurations distances of DNN's local minimum (MD)

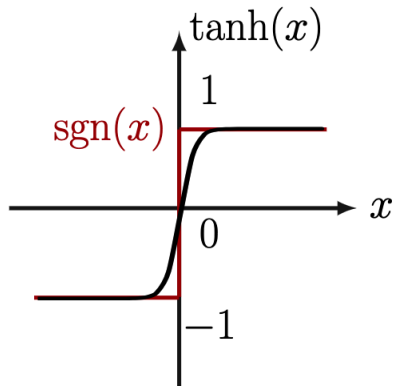
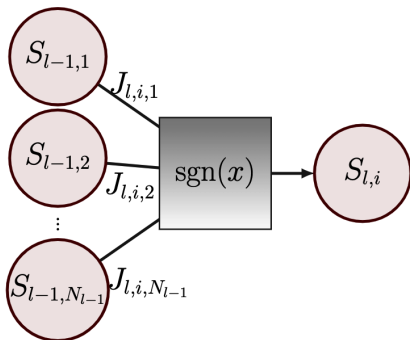


Conclusions and prospects

- Network parameters decorrelate rapidly in *hidden* layers, while showing *slow* relaxations with **aging effects** in layers *near boundaries* due to inputs/outputs.
- Boundary-mediated **liquid-glass transition** phase diagram agrees with prediction based on mean-field calculations using *random inputs/outputs*.

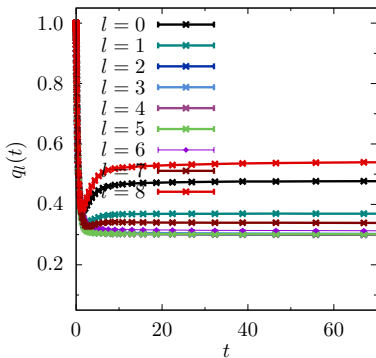
THANKS!

The DNN model: perceptron, activity function.



$q_l(t)$ for DNN with fixed bonds with $\text{ACC} \geq 0.98$

- Similar layer-dependence.



boundary conditions: > 100 ;

$L = 10, N = 3, M = 480, N_{\text{out}} = 2, \beta = 66.7$.